

Solder Joint Inspection on Printed Circuit Boards: A Survey and a Dataset

Furkan Ulger^{ID}, Seniha Esen Yuksel^{ID}, *Senior Member, IEEE*, Atilla Yilmaz^{ID},
and Dincer Gokcen^{ID}, *Member, IEEE*

Abstract—Surface mount technology (SMT) is a procedure for mounting electronic components to the surface of printed circuit boards (PCBs). Although the SMT procedure is more reliable than the conventional through-hole mounting, many errors may occur in SMT lines. In this article, we surveyed methods presented for the optical inspection of solder joints on PCBs. The methods are grouped by problem-solving approach: reference-based, machine learning/computer vision, deep learning, and 3-D reconstruction. We compare and discuss these approaches according to their advantages and disadvantages, with a focus on the more recent deep learning and 3-D reconstruction techniques, and list the public datasets for solder joint inspection. Since defective samples rarely occur in the SMT manufacturing lines, public datasets have few defective samples that are essential for the inspection task. To fill this gap, we publish a public dataset with both normal and different types of SMT errors. Our dataset can be used as a benchmark to compare different algorithms. The dataset is available at <https://github.com/furkanulger/solder-joint-dataset>.

Index Terms—Automated optical inspection (AOI), electronics manufacturing, printed circuit board (PCB) inspection, solder joint inspection.

I. INTRODUCTION

IN CONVENTIONAL electronics assembly lines, printed circuit board (PCB) components were attached through drilled holes in PCBs and soldered to them. This assembly technique is called through-hole technology (THT). However, THT components can take up a lot of space on the PCB and require drilling to insert their leads that incur the cost. With the development in semiconductor technologies, most of the through-hole assembly is substituted by surface mount technology (SMT). The transition from through hole to SMT brought in many advantages, such as reduction in PCB weight, size, and trace lengths. The procedure of placing electronic components in a PCB without inserting them through holes in the board is called SMT [1]. Then, surface mount devices (SMDs) are connected to conductor paths on the PCB by

soldering. During this process, solder errors occur during mass production. However, PCB inspection is challenging since SMDs on PCBs are densely located, and their solder joints can be very small. A tiny error in PCBs can pose fatal errors for their functionality. Detecting these errors in the early stages of production is crucial in increasing production yield, reliability, decreasing rework costs, and scrap rates. There are many studies that discuss SMT procedure and the importance of solder joint inspection [1], [2], [3], [4], [5], [6], [7].

SMT procedure is comprised of the following steps: application and inspection of solder paste, placing electronic components, reflow soldering, and automated optical inspection (AOI) of PCBs. At the first step, a stencil or screen printer is used to apply solder paste. Unique or universal frames are used in printing solder paste for each type of PCB [1]. However, solder paste has to meet stringent quality requirements, for which it is inspected with solder paste inspection (SPI) devices. SPI devices inspect solder paste deposits' shape, slumpness, and viscosity properties. Afterward, electronic components are placed on the soldered pads with a pick-and-place machine. Then, an operator may inspect the PCBs visually against any placement errors. There are many soldering methods for different applications. A widely used one is reflow soldering. In this stage, PCBs are conveyed to a reflow oven for soldering. The solders take their final form by being heated and cooled off in the oven gradually. Thus, solder joints solidify and electronic components become attached to the solder pads permanently. Solvent cleaners are used on the conveyor belt to remove flux residues and contaminants. The SMT procedure is complete with AOI, where both placement errors of the electronic assemblies and solder joints are inspected [1], [8]. An assembled PCB example with solder joints that is tested against errors is given in Fig. 1(a). On the other hand, some electronic package's solder joints are not visible to a top- or side-view camera. These components' solder joints are inspected with automated X-ray inspection (AXI) devices, as shown in Fig. 1(b). A representative PCB assembly line is given in Fig. 2.

For good soldering, there are two utmost important properties: the solder must reach everywhere it is required to without any open joints and the solder must not bridge two different solder joints, which causes a short circuit. These two conditions should be met for good soldering and a PCB to function correctly [9]. Prior to AOI machines, the solder joints were inspected manually by operators. However, manual

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Furkan Ulger is with the Department of Electrical and Electronics Engineering, Hacettepe University, 6800 Ankara, Turkey, and also with Aselsan Inc., 06830 Ankara, Turkey (e-mail: furkan.ulger@stu.ee.hacettepe.edu.tr).

Seniha Esen Yuksel, Atilla Yilmaz, and Dincer Gokcen are with the Department of Electrical and Electronics Engineering, Hacettepe University, 6800 Ankara, Turkey (e-mail: eyuksel@ee.hacettepe.edu.tr; ayilmaz@ee.hacettepe.edu.tr; dgokcen@ee.hacettepe.edu.tr).

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TABLE I
SUMMARY AND COMPARISON OF SURVEYS FOR PCB INSPECTION

Title	Year	Published in	Content
Automated visual inspection: A survey [2]	1982	TPAMI	Image subtraction and template matching algorithms for PCB pattern and chip inspection are discussed.
A survey of automated visual inspection [3]	1995	Comput. Vis. Image Underst.	Template matching, model-based, and hybrid methods for PCB pattern and IC inspection
Automatic PCB inspection algorithms: A survey [4]	1996	Comput. Vis. Image Underst.	In addition to the previous studies, model-based, morphological, and encoding techniques are covered for PCB pattern inspection.
Computational intelligence in mass soldering of electronics—A survey [5]	2012	Expert Syst. Appl.	Traditional machine learning-based classifiers, and shallow neural networks for solder joint inspection.
A review and analysis of automatic optical inspection and quality monitoring methods in electronics industry [6]	2020	IEEE Access	PCB traces, SMD and BGA solder inspection methods based on rules, machine and deep learning, and illumination settings are reviewed.
Recent advances in surface defect inspection of industrial products using deep learning techniques [7]	2021	Int. J. Adv. Manuf. Technol.	Traditional machine learning and deep learning methods for PCB patterns and SMD solder inspection are presented.
Solder Joint Inspection on Printed Circuit Boards: A Survey and a Dataset	2023	Ours	An overview of PCB fabrication, assembly lines, and solder errors with recent deep learning as well as 3D reconstruction based solutions are given. A public dataset is provided for PCB solder joint inspection.

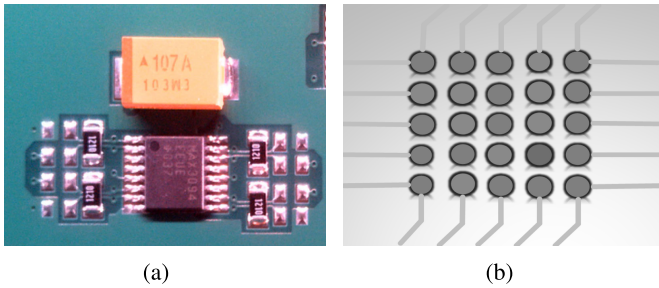


Fig. 1. (a) Assembled board and (b) BGA solder joints' drawing under X-ray.

inspection is both time-consuming and prone to errors. The abrupt development in the electronic production industry entailed automated inspection of PCBs with high accuracy and speed. The AOI machines do not only bring in accuracy and speed but also decrease cost by decreasing the scrap rates and rework. Organizations, such as the Institute of Printed Circuit Boards (IPC), define standards for solder joints. The AOI devices employed in the production lines comply with these standards.

Most of the errors that occur during the SMT process is caused by solder joints. However, solder joint inspection is a difficult problem, as the solder joints under test are generally very small and take on various shapes. Besides, there are tiny differences between a normal and defective solder joint that poses fatal errors in the PCBs [4]. Hence, detecting these errors in the early stages of production is very important.

The approaches presented for solder joint inspection can be mainly divided into four categories: reference-based methods, machine learning (ML)/computer vision (CV), deep learning, and 3-D reconstruction. Reference-based approaches compare the PCB under test with a defect-free version of it. Reference-based methods are easy to implement, and however, their generalization capability is limited so that a defective PCB can only be tested with its defect-free counterpart. Also, images are typically high-dimensional and require learning a complex representation to generalize to new examples. Traditional ML

models are insufficient to learn complicated functions in high-dimensional space. Deep learning models can learn features or representations at increasing levels of abstraction. This enables the model to learn a better representation, hence improving classification accuracy [10], [11]. This also applies to solder joint inspection as elaborated in Section V-C. On the other hand, 3-D reconstruction models solder joints by incorporating additional knowledge with special hardware, which is not available in 2-D approaches.

There are a few surveys from both the earlier and current literature given in Table I by their publication year, venue, and main content. Earlier literature surveys [2], [3], [4], [5] mainly focused on PCB pattern inspection and partially SMD solder joint inspection with mostly reference-based and traditional ML-based approaches. Latterly, deep learning-based methods replaced these methods [6], [7].

Our contributions can be listed as follows compared to the existing literature.

- 1) We give an overview of PCB fabrication materials, assembly procedure, and potential errors to establish a motivation for the inspection.
- 2) We list the public datasets for PCB inspection. To our knowledge, there is no public solder joint dataset with enough defective samples. Our dataset can serve as a benchmark to test algorithms.
- 3) Our dataset includes segmented solder joints that do not require a prior input for feature extraction, model training, and so on.
- 4) Recent deep learning methods with several subareas are covered comprehensively with the application of solder joint inspection.
- 5) An overview of 3-D reconstruction and its application to solder joint inspection is given in detail.

In the rest of this article, we first give an overview of PCB fabrication materials in Section II and solder joint errors and inspection methodologies in Section III. Second, we provide the public datasets in the literature and our dataset that can serve as a benchmark in Section IV. Then, we review the

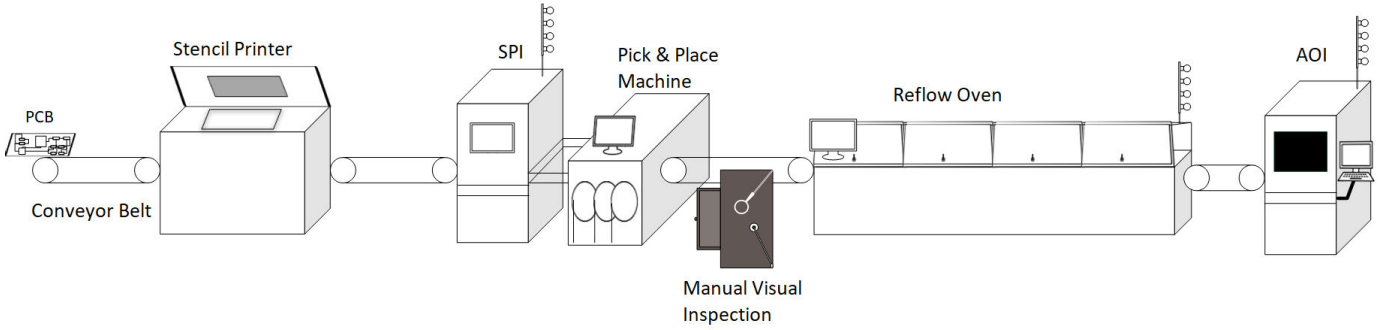


Fig. 2. Electronics PCB assembly line scheme (inspired by [8]).

TABLE II

WIDELY USED LAMINATE TYPES BY THEIR BASE MATERIAL, USAGE ADVANTAGES, AND DISADVANTAGES IN PCB FABRICATION

Laminate type	Strengths	Weaknesses	Use cases
Phenolic laminate	Ease of fabrication, low cost, widely used in the industry	Poor arc resistance, higher water absorption	General purpose applications
Epoxy laminate	High mechanical strength, excellent insulation, low water absorption, excellent adhesion bond	-	General purpose applications
Glass cloth laminate	Satisfactory electrical and mechanical properties for most of the applications, widely used laminate type in the industry	Lower through-hole reliability	Single- and double-sided boards
Prepreg material	Chemical and solvent resistance, higher through-hole reliability	-	Adhesive in between multi-layer board
PTFE laminate	Low moisture absorption, higher peel strength, excellent electrical properties	Low dielectric constant, relatively expensive	Multi-layer boards, RF applications

existing approaches in Section V with a special focus on reference-based, ML/CV, deep learning, and 3-D reconstruction-based methods. These methods are further divided into subgroups according to their prevalence. Then, the approaches are compared and discussed in Section VI by advantages and disadvantages in terms of applicability. Finally, some future directions according to the recent progress in the literature are given to provide a research direction in Section VII.

II. PCB FABRICATION MATERIALS

Prior to conducting a PCB inspection, there are two preceding steps, namely, fabrication and assembly. In this particular section, we aim to provide readers with a concise overview of the PCB fabrication process, with emphasis on the base materials, in order to increase their familiarity with PCB materials.

PCBs are essentially comprised of two parts: base and conductor layer. The base is a thin board made of insulating material, which provides mechanical support to conductors and electronic components. Conductors are typically thin strip copper where the components are attached. The composition of these layers is called copper-clad laminates [12].

Mass production in the electronics industry and increased density of interconnections with smaller packages necessitated multilayer PCBs. A multilayer PCB is produced by stacking up copper-clad laminates and resin layers (prepreg). Interconnection between these layers is ensured by plated through holes. THT components are inserted into these through holes and soldered from the other side, whereas SMDs are directly

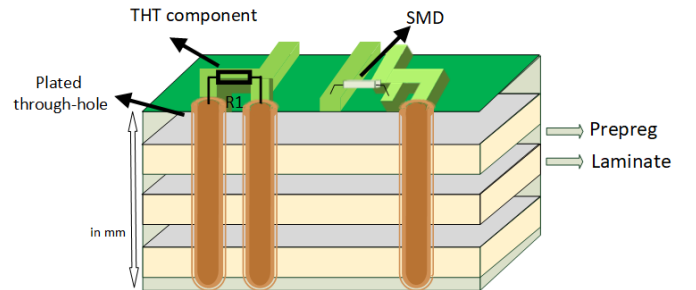


Fig. 3. Multilayer PCB structure with preregs and laminates (inspired by [12]).

attached to the surface of the PCB. PCB laminates consist of reinforcements, resin, and copper foil. The most common reinforcement materials, which provide mechanical strength and rigidity, are paper and glass fiber. The resin holds the laminate together. Copper foil forms the conductive layer on a PCB. Copper foil is most commonly made of copper due to its high conductivity, low resistivity, and lower cost. A multilayer PCB example is given in Fig. 3.

There are many different laminate types that are made of different materials. Each of them has some advantages and disadvantages, and thereby, their use cases can be different. Laminate types used in PCB fabrication are listed in Table II by their strength, weaknesses, and use cases. A widely used laminate type in the industry is phenolic laminate that is formed by phenolic resins reinforced by paper fillers and copper foil is applied on top of it. These laminates have low cost and can easily be fabricated; therefore, they are widely

used in the industry. On the other hand, they have low arc resistance and high water absorption. These properties make them suitable for general-purpose applications. An alternative to phenolic laminate is epoxy laminate, which does not suffer from poor arc resistance and high water absorption. It is a combination of glass/paper fiber filler and epoxy resin. Hence, they are highly durable and excellent insulators. Glass cloth laminates are formed by epoxy resin that is reinforced with glass cloth. Since they provide satisfactory mechanical and electrical properties, they are also widely used in the industry. However, PCBs are exposed to high temperature during soldering, which can expand the glass cloth laminates. The expansion is detrimental to through-hole reliability. Prepreg materials are epoxy resins filled with reinforcements. The materials inherently have high chemical and solvent resistance and are durable at high temperatures, which results in higher through-hole reliability. They are used to adhere laminates of multilayer boards. Finally, polytetrafluoroethylene (PTFE), also known as Teflon, is a thermoplastic material reinforced with glass to form a laminate. The material is relatively expensive, and therefore, they are used where excellent electrical properties are needed such as radio frequency (RF) applications [12], [13].

The PCB fabrication is as important as PCB assembly and inspection for high reliability. High-quality base materials are essential to provide the best electrical and mechanical properties and should be chosen carefully depending on the application. Once the fabrication and assembly processes are complete, the PCB card is sent to the AOI devices for inspection.

III. SOLDER JOINT ERRORS AND INSPECTION

Solder joint errors can be classified into two groups: major and minor defects. Major defects pose fatal errors in the operation of PCBs, whereas minor errors may not cause failures. Major solder joint errors include solder bridges (short), dewetting, solder wicking, solder balls, misaligned components, and tombstoning. Minor errors are cosmetic errors on the solders (discoloration, spots, voids, adhesive contamination, and so on), insufficient, and excessive solder. These errors do not necessarily cause failure in the PCB performance [1]. Some examples of these defective and normal solder joints are given in Fig. 4.

Solder joint errors occurring in an electronics production line are inspected mostly based on optical and X-ray inspection. Optical inspection is carried out by a top-view camera for 2-D PCB imaging. PCBs are either lit by a tiered red–green–blue (RGB) light or coaxial lighting for line scanning. Highlighted solder joint features on the PCBs are used in classification. However, 2-D imaging does not give 3-D information such as height. Therefore, oblique view cameras are used to capture images from different angles. These images are used for 3-D reconstruction of the solder joints, which characterize solder joint errors. The AOI devices can be used to inspect both SMD placement and solder errors that are visible to the cameras.

Some SMT packages, such as ball grid array (BGA) solder joints, are underneath the components; therefore, they are not

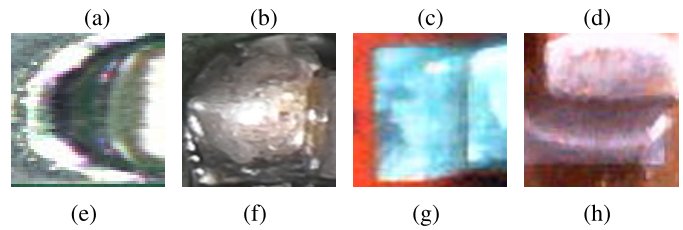


Fig. 4. (a) Normal solder joints of passive SMDs. (b) Shifted component. (c) Normal active SMD solder joints. (d) Shorted solder joints. (e) Sufficient solder. (f) Excessive solder. (g) and (h) Insufficient solder errors from the HU-solder dataset.

visible to side- and top-view cameras. AXI is used to inspect such components. In X-ray inspection, a PCB and its solder joints are scanned by X-ray radiation. Soldered joints absorb these X-rays and become visible as they are made of heavier elements. Meanwhile, PCB and its components are made of lighter elements, so they are more transparent to X-rays. This enables BGA solder joint inspection, especially against voids on the solder [14].

Considering that AOI and AXI are two different solutions and require multiple devices to inspect the different type of SMD packages' solder joints, this incurs costs for PCB assembly lines. Combining these methodologies in a single device would be a good solution to carry out the inspection with a single device.

IV. PCB INSPECTION DATASETS

In this section, we mention the public datasets and related information presented for PCB inspection tasks and introduce our Hacettepe University (HU-solder) dataset. Depending on the type of SMDs, solder joints' shape and size vary. Hence, in order to build a model that can generalize, a lot of normal and defective solder joints are required. Even though there are datasets that include PCB images with normal solder joints, there is no comprehensive dataset that has enough defective solder joints for the inspection task. We provide a public dataset with 2735 normal and 655 defective solder joints. In addition, most of the datasets presented in the literature [15], [16], [17], [18], [19], [20], [21], [22] are labeled PCB component images. These images cover both SMD component body and its solder joints. Since solder joint errors occur on the joint itself rather than the component body, these datasets cannot be directly used in solder joint inspection without prior input. Therefore, we segmented SMD solder joints without covering the component body except for the placement errors where the component body is required. Public datasets presented in the literature are summarized in Table III.

The presented public datasets employ two imaging methods: optical and X-ray computed tomography (CT). RGB-based optical imaging is utilized to capture PCB images and is mainly employed for segmenting or inspecting PCB patterns,

TABLE III
PUBLIC DATASETS PRESENTED IN THE LITERATURE ARE GIVEN ACCORDING TO THE STUDIES, IMAGING METHOD,
RESOLUTION, SIZE OF THE DATASET, AND LABELING INFORMATION

Study	Imaging method	Resolution	Number of PCBs	Labels
[15]	DSLR	4928×3280	165	ICs
Link:	http://www.caa.tuwien.ac.at/cvl/research/cvl-databases/pcb-dslr-dataset/			
[16]	DSLR/ Industrial Camera	NS	48	Resistor, capacitor, LEDs, switch, connector, inductor, IC
Link:	https://sites.google.com/view/chiawen-kuo/home/pcb-component-detection			
[23]	NS	16000×16000	11	Open circuit, short circuit, mouse bite, spurious copper, pinhole
Link:	https://github.com/tangsanli5201/DeepPCB			
[17]	Digital microscope, DSLR	8256×5504	31	Capacitor, resistor, inductor, transistor, diode, IC, text on SMDs
Link:	https://www.trust-hub.org/data			
[18]	Digital microscope, DSLR	NS	165	ICs
Link:	http://homes.soic.indiana.edu/IUPCB-Dataset			
[19]	CCD Camera	2048×2048	NS	Capacitor, IC, resistor, diode, transistor, and SMD text
Link:	https://pan.baidu.com/s/1RjEnSu37l9tPzpEOqEYWsw (password: snqg)			
[24]	16 megapixel HD Camera	4608×3456	10	Open circuit, short circuit, spur, mouse bite, spurious copper, missing hole
Link:	https://robotics.pkusz.edu.cn/resources/dataset/			
[20]	Camera	NS	20	NS
Link:	https://drive.google.com/file/d/1b1GFuKS88nKaH-Nfx2XmhwulUxMwwBA/view			
[21]	DSLR	NS	73	Various SMDs and their text
Link:	https://www.trust-hub.org/#/data/pcb-images			
[22]	NS	NS	NS	Various resistor packages and flipped, missing, tombstone errors
Link:	https://github.com/leiruoshan/PCB-Resistor-Defect-Dataset (in Chinese)			
[25]	CT	9.87 to 32.76um	5	Vias, traces, pads
Link:	https://www.trust-hub.org/#/data/xray-tomography			
HU-solder (ours)	13 megapixel camera	10um, 4196×3104	11	Normal, excessive, insufficient, short solder, and shifted components
Link:	https://github.com/furkanulger/solder-joint-dataset			

NS not stated

identifying SMD placement errors that are observable from top- or side-view cameras. However, since the camera can only capture visible regions, the inner layers of the PCB are not visible. In such situations, X-ray CT imaging is employed.

One of the first public datasets that include PCB images in the literature is presented by Pramerdorfer and Kampel [15]. This dataset consists of 165 PCB images captured by a DSLR camera with a resolution of 4928×3280 . Only active SMDs such as integrated circuits (ICs) are labeled on the PCBs. These labels are used in detecting ICs on PCBs. Reza et al. [18] similarly labeled ICs to detect them. Resistor, capacitor, LEDs, switch, inductor, connector, and ICs on 48 PCBs are labeled

in [16]. Lu et al. [17] captured PCB images with resolution of 8256×5504 . They labeled IC, capacitor, resistor, inductor, transistor, diodes, and text on top of SMDs to classify them. Likewise, capacitor, resistor, IC, diode, transistor, and texts are labeled in [19]. Labeled SMDs are used in finding placement errors of SMDs. In addition, text on detected SMDs is used in finding spelling errors. Jessurun et al. [21] labeled a variety of SMDs and their text on 73 PCB images. Different than the other studies, polygon labeling is used instead of rectangular labeling for more precise segmentation. On the other hand, Fridman et al. [20] published PCBs in pairs as reference and defective version of the reference. The difference

TABLE IV
OVERVIEW OF THE PRINCIPLES, ADVANTAGES, DISADVANTAGES, AND APPLICATIONS OF THE
METHODS IN THE TAXONOMY FOR SOLDER JOINT INSPECTION

Principle	Advantages	Disadvantages	Applications
Image subtraction	Trivial to implement	Requires many templates, robust registration, and susceptible to environmental changes	Pixel comparison of the reference and PCB under test
Template matching	More robust than image subtraction	Requires many templates, computationally expensive	Correlation/feature comparison of the reference and PCB under test
Machine learning/ computer vision	Achieves high accuracy, reliable with distinctive features	Requires special hardware for color highlight, feature engineering	Segmentation and classification of solder joints, SMDs, PCB traces
Deep supervised learning	Achieves high accuracy, can perform localization and classification with a single model.	Requires a lot of labelled normal and defective samples	Segmentation and classification of solder joints, SMDs, PCB traces
Deep unsupervised learning	Does not require labelled defective samples in training	Lower accuracy compared to supervised learning	Anomaly detection of solder joints
3D reconstruction	Facilitates inspection by preserving 3D information of solder joints. Widely used in the industry.	Requires more time and special hardware	3D modelling of solder joints

between these pairs is used to detect errors. Wu et al. [22] annotated over 260.000 resistor packages with various sizes. The resistor errors include flipped, missing, and tombstone errors. These components are generally rotated, and hence, horizontal/vertical bounding box annotations are not precise enough. The authors provided oriented bounding box annotations in addition to the horizontal ones.

Other than SMDs on PCBs, a few studies share datasets with PCB traces on bareboards [23], [24]. Different bareboard defects, such as missing hole, mouse bite, open circuit, short circuit, spur, and pinholes, are labeled. Hence, the defects that occur on PCB traces can be detected and classified.

The majority of the studies utilized optical imaging to inspect or segment the visible parts of PCBs. However, optical imaging has limitations as it can only capture the regions visible to the camera. To overcome this, Mehta et al. [25] released a publicly available X-ray PCB image dataset captured by CT of five PCBs at varying resolutions. X-ray imaging can penetrate concrete surfaces, which makes it possible to ensure the inner layers of the PCBs. To achieve this, PCB vias, traces, and pads in the inner layers were annotated.

The aforementioned public datasets can be used in the detection and inspection of SMDs on PCBs. However, PCB errors occur rarely on the production lines or it takes time to create one manually. Therefore, there are not enough defective solder joints/SMDs on the public datasets. This makes PCB inspection task hard with these datasets.

HU-Solder: To address this gap, we have released a dataset featuring segmented normal and defective solder joints/SMDs from PCBs. The dataset consists of 92 excessive, 149 insufficient, 300 shorted solder joints, 114 shifted components, and 2735 normal solder joints. Our dataset is unique in that it was collected with multiple AOI devices. Initially, the PCBs were scanned and captured using a top-view camera with a resolution of 4196×3104 optically. Our dataset differs slightly from others in that the solder joints in the captured PCB images are labeled with their class labels and bounding box coordinates using [26] to identify the regions of interest. Finally, the labeled solder joints are segmented for the final inspection. Examples of normal and faulty SMD solder joints

in the dataset are given in Fig. 4. To our knowledge, there is no public dataset with 655 defective solder joints, which is essential for the inspection task to build a model. In addition, our dataset includes segmented solder joints; therefore, it can be used in training/deploying a model without prior input for solder joint inspection and can be used as a benchmark to compare different algorithms.

V. METHODOLOGY OF TAXONOMY

Fig. 5 presents the methodologies used in the solder joint inspection taxonomy. The taxonomy is comprised of four primary divisions, namely, reference-based, ML/CV, deep learning, and 3-D reconstruction, along with their respective subcategories. The main divisions and subcategories are defined by the problem-solving approach utilized for the solder joint inspection task. An overview regarding the methods in the taxonomy is summarized in Table IV by their advantages, disadvantages, and applications. We discuss the knowledge gap and possible solutions at the end of each section.

A. Reference-Based Methods

Reference-based methods compare a PCB under test with a defect-free (reference) version of it either pixelwise or featurewise [4]. In essence, image subtraction and template matching methods are employed in these approaches, which are detailed in the following.

1) *Image Subtraction*: A reference PCB image and a PCB under test are compared pixelwise in image subtraction. The reference version is a defect-free version of the PCB under test. The subtracted image shows the difference between the reference and test images, which are called defects. An example of the logical XOR operation used in image subtraction is given in Fig. 6. Reference-based methods are both employed in bareboard inspection [27], [28], [29], [30] and assembled board inspection [30], [31], [32], [33]. In order to carry out pixelwise comparison, segmentation and image registration algorithms are used for preprocessing. Then, the corresponding pixels are subtracted from each other to detect errors [27],

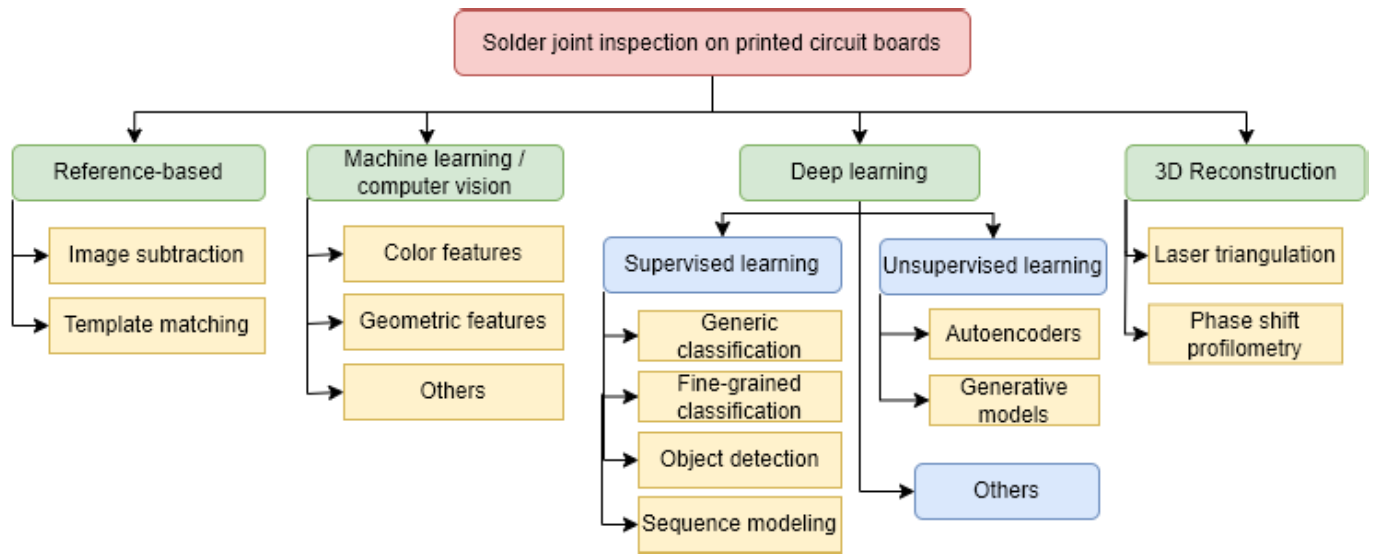


Fig. 5. Taxonomy of methods reviewed in this survey and proposed for optical inspection of PCBs.

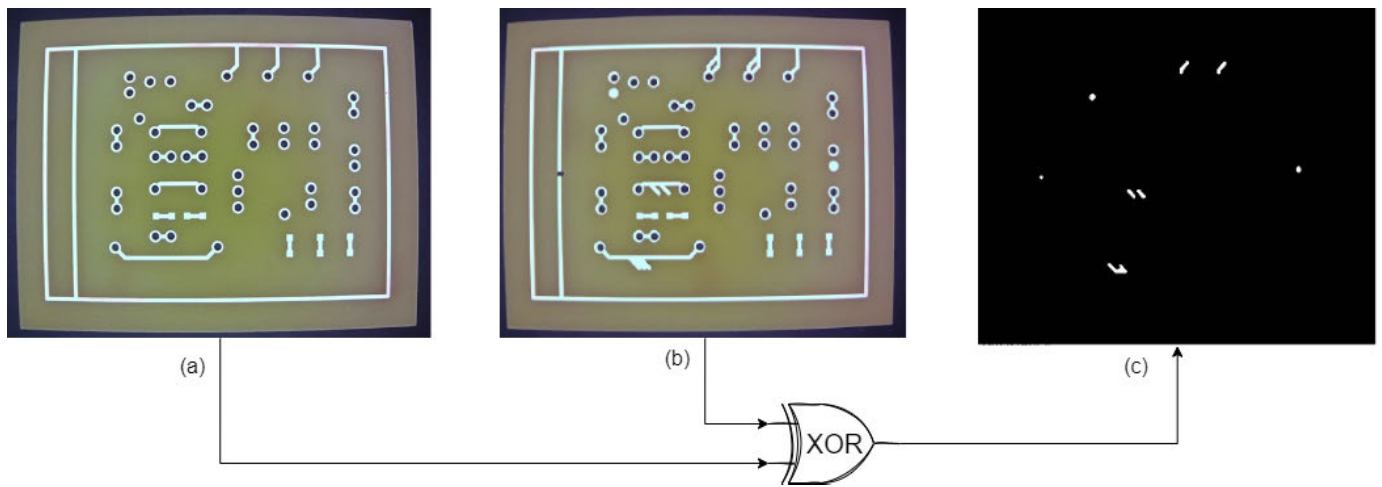


Fig. 6. Subtraction operation between (a) normal and (b) defective PCB traces with a logical XOR operation [30] yields (c) defective PCB traces.

[30], [33]. Background subtraction utilizing logical operations fails to account for environmental changes, which can result in false alarms. To address this issue, background modeling techniques are employed to model reference PCB backgrounds. Xie et al. [34] used average intensity values, mean, and variance of gray values of passive SMD solder joints to model normal samples. Then, the difference between the defective and normal solder joints is computed to detect errors. Instead of selecting features manually, Sundaraj et al. [32] modeled the reference PCBs with Gaussian mixture models (GMMs). The distance between the test image and the modeled distribution yields the placement errors of SMDs on the PCB. The image subtraction method is not restricted to the spatial domain. Tsai and Huang [35] showed that the subtraction in the spatial domain can still cause some deviations in the subtracted image even after precise alignment. This can prevent detecting subtle errors. Hence, they employ Fourier transformation on the PCB images to analyze the difference in the frequency domain, due to its robustness against translation and noise.

Remarks: Reference-based methods with image subtraction are easy to implement; however, they are susceptible to illumination conditions, color, and reflection changes. They also rely on image registration to compare the corresponding pixels. This confines their practical usage. On the other hand, these methods assume that the components used in the reference and test PCBs are visually identical. However, SMDs with the same functionality can have different shapes and colors, such as aluminum and tantalum capacitors. In these cases, pixelwise comparison of PCB image pairs can cause false alarms. This problem can be alleviated by doing comparisons with multiple references.

2) *Template Matching:* Template matching-based methods extract and compare features or correlations. The earliest methods [36], [37], [38], [39] applied to PCB inspection extract features from a test image and compare it with the reference as surveyed in [2] and [4]. According to the degree of the deviation of features from reference, the test image is evaluated as normal or defective. Later, the correlation between

TABLE V
SOLDER INSPECTION METHODS PROPOSED IN THE LITERATURE ACCORDING TO THE TYPE OF ERRORS,
TIERED RGB ILLUMINATOR USAGE, PROCESSING METHODS, FEATURES, AND CLASSIFIERS

Study	Solder errors	RGB	Processing methods	Features	Classifiers
[59]	Normal, insufficient, excessive, missing solder, and dewetting	✓	Windowing, thresholding, run-length codes	Area, perimeter, circularity, center of mass	Decision tree
[52], [53]	Normal, pseudo, shifted and rotated component, insufficient solder, tombstone	✓	Entropy [53], genetic algorithm [52]	Average gray values, percentage of highlight, correlation	SVM, Bayes [53], MLP [52]
[72]	Excessive, insufficient, missing, normal solder	✓	NS	Average gray values, percentage of highlight	SVM
[54]	Normal, missing, rotated, pseudo and excessive solder	✓	subregion division, Adaboost	Color area, weight, continuity ve contrast ratio	Binary decision and regression tree
[124]	Normal, missing, and excessive solder	✓	Cross and auto-correlation	Highlighted areas and locations	MLP
[64]	Shifted, insufficient, missing, pseudo solder, tombstone, wrong polarity SMDs	✓	Subregion division	Placement, logic and shape features	NS
[62]	Normal, pseudo, missing, excessive solder, missing and shifted component and tombstone	✓	Subregion division	Solder area and distribution, center of mass, color changes	LVQ
[67]	Insufficient, excessive, missing and normal solder	✓	Subregion division	Average gray intensity, percentage of highlight, surface area and curvature	k-Nearest Neighbour (kNN)
[55]	Normal, wrong, missing, rotated, tombstone, shifted, excessive, pseudo solder	✓	Genetic algorithm	Average gray values, percentage of highlight	SVM
[70]	Normal / defective solder	-	Segmentation, thresholding	Solder area, perimeter, variance	MLP
[39]	Missing, misaligned, rotated, wrong SMD, bad solder	-	Brightness and size normalization, windowing	Normalized standard deviation and volume, pixel intensity of gray values	NS
[41], [44]	Missing, misaligned, rotated SMDs	-	Corner detection, genetic algorithm [41], Discrete Cosine Transform [44]	Correlation [44]	-
[30]–[32]	Placement errors	-	Segmentation, image registration [30]	Image subtraction	MLP [31]
[49], [125]	Normal, insufficient, excessive solder	-	Illumination normalization, segmentation	Discrete Cosine Transform, Gabor, Wavelet features	Mahalanobis distance
[77]	Normal, insufficient, excessive	-	Multi-angle view	-	LVQ
[56]	Pseudo, insufficient, missing, shifted solder, short circuit, tombstone, bent lead	✓	Segmentation	Shape, digital, logical features	NS
[68]	Normal, insufficient, excessive solder	-	Segmentation	Geometric, wavelet, correlation	MLP
[75], [76]	Normal, insufficient, excessive solder	-	-	-	LVQ [75], [76], Fuzzy rule-based classifier [76]
[60], [61]	Normal, missing, insufficient, excessive solder	✓	Thresholding, segmentation [61]	Average gray values [61], percentage of highlight, slope angle [60], [61]	MLP, Bayes classifier [61]
[81]	Normal / defective solder	-	Dimension reduction, YIQ color space	Color features	k-NN
[126], [127]	Normal / defective solder	✓	HSV color space	Mixture of Gaussian [126], background subtraction [127]	Frequency map
[80]	Normal / defective solder	✓	HSV color space	Location information	Thresholding
[33]	Normal / defective solder	✓	Adaptive template production, HSV color space	Image subtraction	Weighted pixels

^{NS} not stated

test and reference images is also compared to find errors. Tsai et al. [40] calculated normalized cross correlation (NCC) between reference and test images, where lower correlation

values indicate the presence of errors and higher correlation values indicate defect-free regions. After calculating the correlation, smoothing filters are applied to alleviate false alarms.

TABLE VI
DEEP LEARNING-BASED METHODS IN THE LITERATURE FOR SOLDER JOINTS INSPECTION ARE GIVEN BY THE
ERRORS TO BE INSPECTED, TIERED RGB ILLUMINATOR USAGE, AND PROCESSING METHODS

Study	Solder errors	RGB	Processing methods	Model
[84]	Defective / normal	✓	Image tiling	CNN
[85]	Defective, normal, shifted, tombstone, missing SMD	✓	-	Mask-RCNN
[94]	Excessive/insufficient solder, short circuit, voids, debris	-	PCB segmentation and parcellation	Faster RCNN, YOLO, FPN
[58], [108]	Defective / normal	✓	Active learning [58]	YOLO, CNN
[22]	Flipped, missing, tombstone errors	-	Denoising, color transformations, registration, mark hole detection	MobileNet based object detector
[101]	Defective / normal	✓	Denoising, normalization, color transformation	Linear Autoencoder
[103]	Defective / normal	-	NS	Adversarial Autoencoder
[86]	Defective / normal	✓	Binary image subtraction, thresholding	GAN, encoder
[100]	Defective / normal	-	Reconstruction error	VAE
[102]	Defective / normal	-	-	Multi-task learning
[57]	Defective / normal	-	Segmentation, thresholding, anomaly score	VAE
[87]	Excessive and insufficient solder, Manhattan, missing, normal, tombstone	✓	Subregion division	CNN
[97]	IC and capacitor errors	-	ROI localization under a thermal camera	CNN, autoencoder, YOLOv5
[95]	Missing hole, mouse bite, spur, open circuit, short, spurious copper	-	-	Feature fusion, FPN
[96]	Short, miss hole, spurious copper, spur, mouse bite, open circuit	-	Focal loss	FPN
[105]	Placement errors	-	Geometric transformations, MSE	Reinforcement learning
[92]	Defective / normal	-	Image concatenation	3D-CNN, LSTM
[98]	Defective / normal	-	Localization with YOLACT	Entropy regularization with CNN
[16]	SMD types	-	Localization with Faster R-CNN	Graph network, triplet loss
[102]	Defective / normal	-	Geometric features	Multi-task learning
[92]	Defective / normal	-	Feature concatenation	3D CNN and LSTM

^{NS} not stated

Normalized cross correlation and correlation coefficient are used in finding placement errors (missing, misaligned, and rotated SMDs) [41], [42], [43], [44]. Searching the whole PCB image with a template image is time-consuming and computationally expensive. To decrease these computation requirements, several alternatives have been proposed. Search

space is constrained by employing a Canny edge detector and genetic algorithm in [41]. Search time is reduced by a particle swarm optimization in [42]. Correlation is calculated for 1-D feature descriptors instead of images to decrease computations in [44]. Discrete wavelet transformation is applied before template matching to decrease computation time and storage

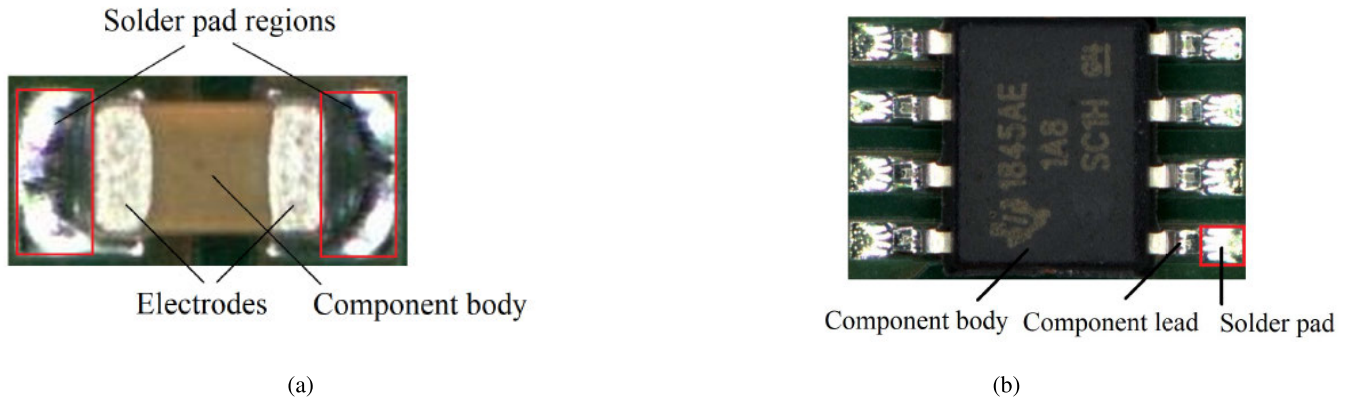


Fig. 7. SMD parts [57]. (a) Passive SMD. (b) IC component (active SMD) parcellation.

in [43] and [45]. Furthermore, Wang et al. [46] thresholded the image before calculating the correlation coefficient to decrease false alarms.

Remarks: Template matching is more robust than image subtraction and it is less susceptible to environmental changes. However, template matching-based methods require infinitely many templates to describe all possible references. In addition, searching a template in a large image is computationally expensive and slow. These restrict the use in practical applications.

B. ML-/CV-Based Methods

ML-/CV-based methods are mainly divided into three phases: preprocessing, feature extraction, and classification [47]. The preprocessing methods are used to facilitate the segmentation of region of interest (ROI). After obtaining the ROI, predefined features are extracted from these regions. Finally, extracted features are used in characterizing errors for classification. Traditional ML-based classifiers are trained with these features to classify errors. The ROI can be a solder joint, SMD body, PCB traces, and so on depending on the inspection task. The methods based on ML/CV are given in Table V.

Segmenting solder joints on a PCB is a crucial step for correct inspection. Many methods are applied before segmentation to obtain the solder joints more accurately. Binary thresholding and filtering are applied to distinguish solder joints from the PCB background in [48]. Binary thresholding is applied to separate these regions in RGB space. After thresholding, background noise is removed by filtering. Mar et al. [49] used Hough line transformation to align PCBs and illumination normalization, which eliminates the low-frequency discrete cosine transform coefficients to ensure the uniform distribution of the illumination over a PCB. They find that the Y component in the YIQ color space account for most of the information to segment these regions. Therefore, they convert the RGB solder joint images to YIQ color space and threshold it. This enables more accurate segmentation. However, these methods are only applicable to bareboards since assembled PCBs have many different parts in different intensity values, such as component body, markings, text, patterns, and vias. Gray pixel values are not enough to distinguish them from

each other. To provide robust segmentation of solder joints on the assembled PCBs, Zeng et al. [50], [51] proposed utilizing the specularity of the solders. A tiered RGB light is employed to highlight solder surfaces to distinguish them from SMDs, PCB traces, and other irrelevant regions that do not have a specular surface. Color and illumination distribution are used over the highlighted solder joints of both IC and passive SMDs. First, reflective regions on the solder joints are detected by thresholding. Then, holes and signs on PCB are filtered by using the color distribution modeled by GMM [51].

It is important to extract features from the distinctive soldered regions for accurate inspection. ICs are mainly comprised of component body, lead, and solder pads. In addition, passive SMDs have electrodes, as shown in Fig. 7(a) and (b). Features are only extracted from the component body and its solder pads [52], [53]. Both IC and passive SMDs are divided into further subregions and different features are extracted from different subregions [54], [55], [56] to detect more type of solder errors. Song et al. [55] proposed using a genetic algorithm to locate where to extract features on the electronic components automatically.

To classify solder joints, numerous features are utilized; however, not all features are equally significant. Feature selection mechanisms, such as Adaboost [54], information gain [53], and genetic algorithm [52], are implemented to identify discriminative features and enhance classification accuracy. In the Adaboost algorithm, weak classifiers are designed for each feature and the feature value of the samples is computed. The features with the lowest classification errors are selected [54]. Information gain is determined by calculating the entropy for each feature to determine its usefulness [53]. In a genetic algorithm, input features are encoded as random binary values, referred to as populations, which are a set of possible solutions. Each attempted solution is optimized over the mean squared error between the prediction and ground truth to approximate a good solution. The best solutions are combined to generate a new population until a satisfactory solution is obtained [52].

It should be mentioned that labeled data are difficult to obtain and requires a lot of labor. Typically, thousands of labeled samples are required to train a classifier that can generalize well. One can resort to active learning, where a part

of the dataset is labeled, which is used to train the classifier as in [58]. Then, the trained model is used to label the rest of the dataset. This facilitates the labored labeling procedure; however, the model trained with few samples can misclassify, and hence, the labeled data should be double-checked.

ML-/CV-based methods aim to find features that are helpful in distinguishing normal and defective solder joints, PCB traces, or SMDs. These features can be grouped as color, geometric, and others. They are generally extracted from solder joints and component body and used in a classifier to determine whether a solder joint is normal or defective, and the type of the defect.

1) *Color Features*: Capson and Eng [59] used blue and red tiered illumination to create color contours on the reflective solder surface for solder joint inspection. Thereby, 3-D solder information can be obtained in two dimensions. The tiered RGB illumination is reflected from the specular solder surface by different angles, and hence, different color contours occur for different types of solder joint errors [60]. The color features extracted from the color contours as a result of the illumination are mainly average gray values, percentage of highlight [52], [53], [55], [61], and color changes [62]. In case the color features are not distinctive, additional features, such as color changes, are required for classification [62]. The RGB illumination is widely used to create the color contours, which characterize errors, in industrial AOI devices as well. When compared to the RGB color space, Fonseka and Jayasinghe [63] determined that hue–saturation–value (HSV) color space was more appropriate for segmenting solder joints from the PCB background based on the mean and standard deviation ratio.

2) *Geometric Features*: Different SMDs have different solder surface shapes. Hence, geometric information can be an effective way to ease classification. Geometric features are commonly used along with color features in the literature. These are mainly the solder area [62], [64], [65], perimeter, length, height, width [59], [66], polarity area [64], solder curvature, and inertia [67]. In addition, color distribution, center of mass, and pixel continuity on solder joints of IC lead and solder pads are used in the solder error classification [56], [62], [64]. Hongwei et al. [54] utilized the area, continuity, and contrast in each color channel.

As opposed to the widely used RGB illumination, some studies aim to carry out the inspection without using a specialized illumination [49], [68], [69], [70]. Chiu and Perng [69] lightened passive SMD solder joints with a single-tier circular illuminator. Percentage of the highlighted regions [69], [70] is used in the classification. It is shown that the location of bright solder regions can change with the type of solder error. They characterize solder errors by the change in the curvature of these bright regions.

3) *Others*: Different features are extracted with various transformations. Acciani et al. [68] utilized the correlation between wavelet features of reference and test images. Wavelet transform creates a multiresolution pyramid of images. As a result of the wavelet transform, approximation and vertical coefficients are evaluated for each type of solder error and compared with those of the reference solder joint image.

Mar et al. [49] showed that Gabor features are more useful than wavelet features since the log-Gabor filter can capture visual properties, such as spatial localization, orientation, selectivity, and spatial frequency characteristics. However, they are susceptible to degradation or distortion of the images. These methods do not require specialized illumination while enabling inspection with high accuracy. However, these methods require more time and are computationally more expensive than using predefined features.

After feature extraction from the ROI, the extracted features are processed in a classifier to classify solder joints/PCB traces/SMDs. The studies in the literature employ several classifiers: multilayer perceptron (MLP) [31], [52], [61], [68], [70], [71], naive Bayes classifier [53], [61], support vector machine (SVM) [53], [55], [72], decision tree [54], random forest [73], K-Means [74], and learning vector quantization [62], [75], [76], [77], [78]. Some studies [53], [61] employed multiple classifiers to improve classification accuracy further. Random forest is shown to outperform k-NN, decision tree, SVM, and MLP in the classification [79].

Robust principal component analysis (PCA) enabled learning a low-dimensional representation of solder joints [80]. This representation enabled higher classification accuracy; however, calculating the eigenvalues of large matrices is computationally expensive. Multiple templates are constructed by a fast iterative shrinkage-thresholding algorithm (FISTA) that can both generalize better and is computationally efficient [33].

On the other hand, Jiang et al. [81] suggested the use of manifold learning to create submanifolds that comprise biological color feature maps for SPI. In essence, manifold learning involves learning a linear mapping to obtain a lower dimensional representation for classification. This is accomplished through a discriminative analysis method that minimizes the distance within the class and maximizes the distance between classes. Subsequently, test images are projected onto the learned submanifold and classified based on their proximity to the good quality or defective group.

Liao et al. [82] computed the Kullback–Leibler (KL) divergence between probability distributions of the reference PCB image and test image to measure the defectiveness of PCB traces. Thereby, the inspection can be done without using the widely used RGB illumination mentioned in Section V-B.

Remarks: ML/CV-based methods are robust and accurate methods with distinctive features for solder joint inspection. Color features are generally found more useful than the geometric and texture features [81]. On the other hand, the color features typically require tiered RGB illumination to highlight solder surfaces. Solder joint errors are determined by these highlights. However, how these highlights are achieved in practice with illumination is a gray area. In order to implement the color highlight methodology, it is critical to know the position and angle information of each RGB light with respect to the solder joints. Besides, end-to-end inspection with ML/CV requires localization of the solder joints before feature extraction, which is not a trivial task. On the other hand, the difficult feature engineering process can be automated by learning a good representation, various transformations, and optimization that make it easier to extract useful information.

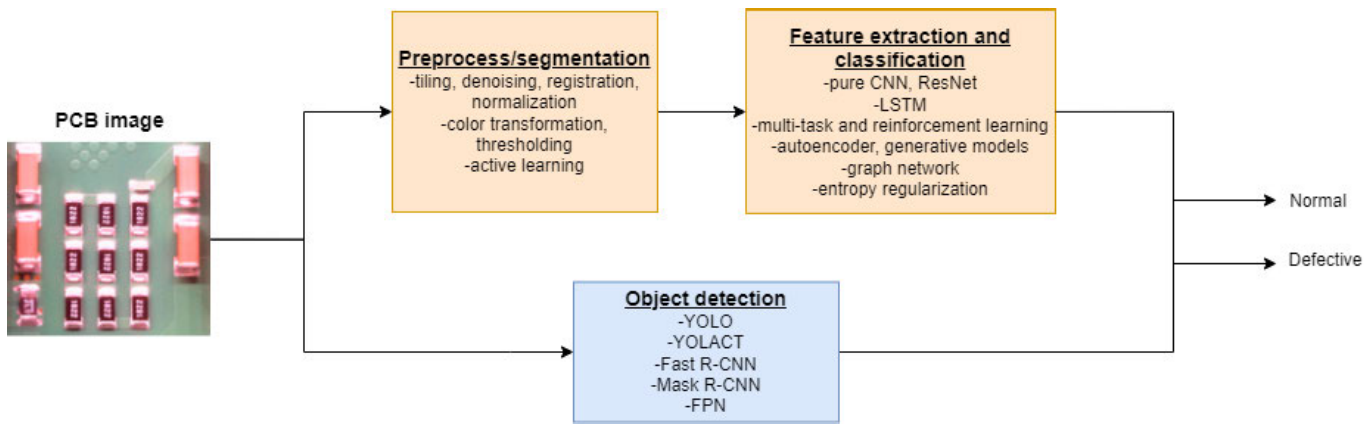


Fig. 8. Typical pipeline for PCB inspection with deep learning.

C. Deep-Learning-Based Methods

In recent years, it is shown that deep neural network based approaches outperformed the traditional ML- and CV-based approaches [83]. This also applies to the PCB inspection task. Deep neural networks achieved higher accuracy across different solder datasets compared to classifiers, such as Bayes classifier, SVM, MLP, PCA, and GMM [84], [85], [86]. Deep learning-based methods are more advantageous since they provide end-to-end learning and can learn both high- and low-level features, which does not require engineering features by hand. Considering that the success of PCB inspection task mostly relies on these features, many deep learning-based methods have been applied to this task as well. These methods are summarized in Table VI.

A typical pipeline for PCB inspection using deep learning is given in Fig. 8. Object detectors allow a single-step localization and classification of errors on a PCB image, whereas the other methods require localization/segmentation of the ROI first and then the feature extraction and classification.

Deep learning-based methods applied to PCB inspection can be mainly grouped under supervised, unsupervised learning, and others.

1) Supervised Learning:

a) *Pure convolutional neural networks:* Convolutional neural networks (CNNs) consist of layers of neurons, the parameters of which are typically optimized over an objective that minimizes cross entropy between the model distribution and distribution over the labels. Cai et al. [84] designed a dual-stream CNN architecture to inspect active SMD (ICs) solder joints where one stream is used to localize where the probable error is and the other is used in classification. Kim and Park [87] used dual-stream CNNs on passive SMD solder joints and merged the extracted features for the final classification. They show that the dual-stream CNN-based approach achieves higher accuracy compared to the single-stream, modeling-based, and traditional ML-based methods. As the network gets deeper, these models suffer from vanishing gradients that stop learning since the gradient with respect to model weights is multiplied at each layer. By using residual skip connections between layers, some layers can be skipped during backpropagation. This can mitigate the vanishing

gradient problem and increase generalization. Hence, instead of using a CNN, ResNet-50 is employed in [88], which improves the classification accuracy. In an electronics manufacturing line, PCB errors occur rarely, and therefore, there can be many more normal samples than the defective ones in a dataset. In line with this, Zhang et al. [89] proposed to minimize a weighted cross-entropy loss that assigns more weights to the defective samples. This approach decreases the number of false alarms.

b) *Sequence modeling:* Sequence models are designed to process sequential data that depend on each other and are indexed by time steps. Since there is a dependency, these models take information from a previous state to the current state unlike the traditional deep neural networks (NNs). Solder paste data include locations of solder paste and solder paste printer's temporal information [90]. Hence, both temporal information and spatial information are used continuously to inspect solder paste during solder paste printing, which is not possible with only the spatial information. Convolutional long short-term memories (LSTMs) and spatiotemporal attention mechanisms are used to model normal solder paste printing data. LSTM is comprised of a chain of repeating four neural networks interacting with each other. There are several gate layers in LSTM: forget gate determines whether to keep the current value in the memory, the input gate is used to store the new information, and a weighted combination of the previous and current state of the memory is given at the output layer [91]. The attention mechanism is used to overcome the long-term dependency problem caused by too many hidden states. The model trained with normal data cannot reconstruct anomaly map for defective samples. The degree of error is determined by comparing the reconstruction error between the input and reconstruction in test. The sequence modeling approach outperforms standard classification with the CNN that uses only spatial information.

On the other hand, Zhang et al. [92] performed an LSTM-based inspection of X-ray BGA images. First, each image feature is extracted by a CNN. Then, these features are concatenated and classified by an LSTM module that is followed by fully connected layers. This approach is compared with classifying with CNN on 2-D and 3-D images. The authors

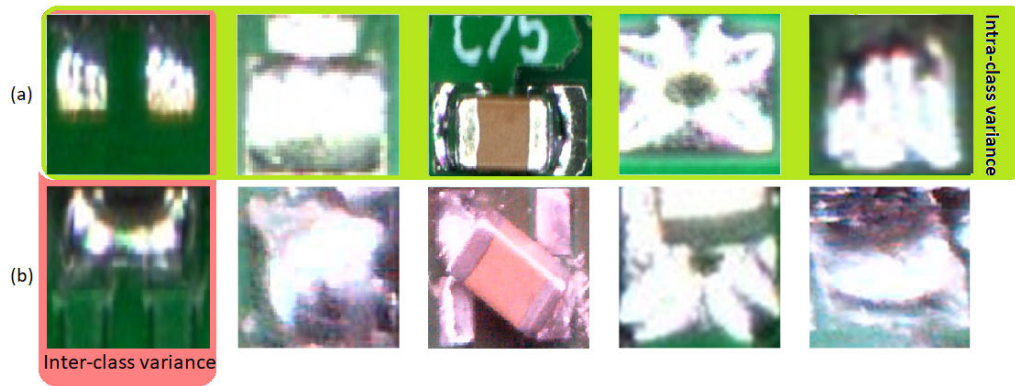


Fig. 9. (a) Normal and (b) defective solder joints are given in the first and second rows, respectively. Normal and defective samples are highly similar; hence, they exhibit low intraclass variance, whereas samples from the category are visually very different (high interclass variance) [98].

found that the proposed method is on par with the LSTM-based model.

Shen et al. [19] employed an LSTM-based CNN architecture to carry out optical character recognition (OCR) on the text of SMDs to find wrong component errors. Localized SMDs' text features are converted to label sequences by a bidirectional LSTM module. The final predictions are made by gated recurrent units. The proposed sequence modeling pipeline outperformed other deep learning-based OCR methods on SMD text recognition.

c) Object detectors: They aim to detect the location of an object in an image and determine which category it belongs to. The detectors are mainly based on region proposals or classification/regression. Region proposal-based ones include two steps: propose regions, then classify into object classes, and find localization coordinates, whereas classification/regression-based approaches, such as you look only once (YOLO) [93], can directly map image pixels to object classes and do localization.

Wu et al. [85] used mask region-based CNN (R-CNN) to both localize and classify solder joints, which outperformed traditional approaches based on SVM, ensemble learning, and PCA. Since supervised models typically require a lot of samples to learn from, Dai et al. [58] resorted to active learning that uses the model to annotate unlabeled samples and reduce manual labeling labor. A YOLO detector is used to localize ROIs. Multiple object detector algorithms, such as Faster R-CNN and YOLO, are used in tandem to further improve localization and classification accuracy [94].

PCB defects are very small compared to generic objects, and therefore, object detectors have difficulties to localize and classify very small objects. Yu et al. [95] extracted features from a feature pyramid network (FPN) that yields feature maps at different scales, and then, the features are fused to get a richer feature representation. Finally, the receptive field for the target is adjusted adaptively to detect PCB trace errors. Meanwhile, a focal loss is used to improve the classification performance of hard to distinguish samples for classification and localization with an FPN [96]. Both of these methods outperformed YOLO and Faster R-CNN on different backbones.

Deep learning models have too many parameters that make model training computationally expensive, but PCB

inspection requires high speed and is generally carried out under restricted computational resources. Therefore, Wu et al. [22] used a MobileNet architecture, which has fewer parameters with fused convolutional layers for faster and computationally less expensive localization of SMD placement errors.

Visual information under visible light is not helpful in detecting SMD errors caused by internal characteristics. Therefore, YOLOv5 is applied to localize and classify IC and capacitor's internal errors from thermal PCB images that can reveal internal characteristics of components [97].

d) Fine-grained classification: Another approach to the inspection task can be fine-grained image classification (FGIC). The problem is challenging so that normal and defective PCB components/solder joints can be visually very similar (low interclass variance), whereas same classes can be very different visually (large intraclass variance), as shown in Fig. 9. Hence, the inspection task can be considered an FGIC problem as in [16] and [98]. In FGIC, the aim is capturing subtle visual differences such as discriminative features to distinguish different classes [99].

Kuo et al. [16] proposed a pipeline to classify fine-grained PCB components. The components are localized with Faster R-CNN. Features are extracted by a graph network, and a triplet loss is used to classify the components. Ulger et al. [98] showed that solder joints are fine-grained by comparing their feature diversity to a generic dataset. An instance segmentation algorithm, YOLACT, is used for the segmentation of solder joints. Then, an entropy regularizer based on skewed Jensen–Shannon divergence is used for fine-grained classification of solder joints to penalize overconfident model predictions.

Remarks: To sum up, different supervised learning-based methods are effective under different circumstances. For instance, if temporal data are available, sequence models can be employed to ensure continuous inspection. Solder joints are located on different parts of PCBs, and therefore, they cannot be directly inspected over a PCB image. Object detectors enable both localization and classification of solder joint errors across various parts of the PCBs. Moreover, a loss function can be designed specific to this task such as FGIC. Supervised learning methods can achieve high accuracy by leveraging the labeled data, but they demand a large amount of labeled data

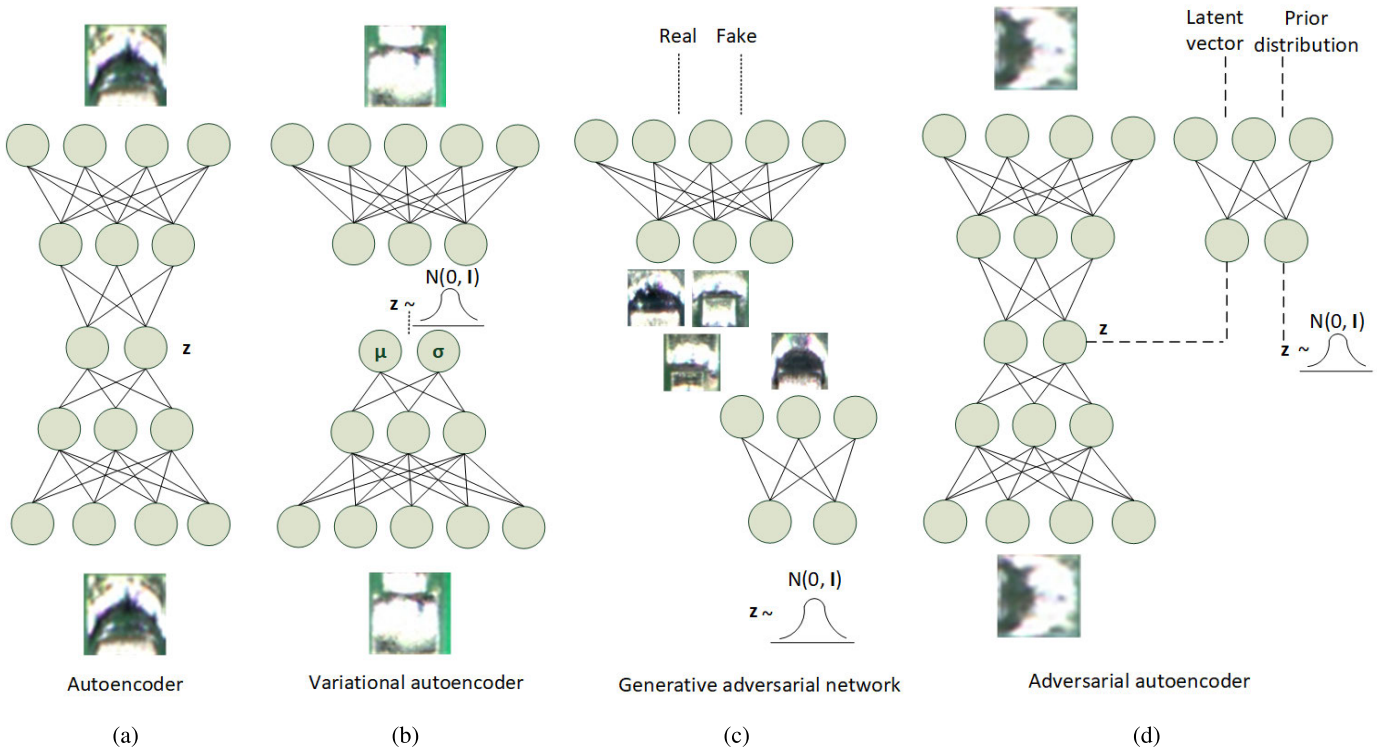


Fig. 10. Latent variable models: (a) autoencoder, (b) VAE, (c) GAN, and (d) AAE.

for both normal and defective samples. Since defective data are scarce in solder joint inspection, developing a supervised model that can generalize well without sufficient data can be challenging. In these cases, semisupervised learning, unsupervised learning, transfer learning, and few-shot learning can be resorted at expense of model accuracy.

2) *Unsupervised Learning*: In supervised learning with neural networks, both normal and defective solder joints are used. However, solder joint errors occur rarely on a production line. Besides, solder joints can take on various shapes. Therefore, an extensive dataset is required to cover most type of solder joint errors. Due to that fact, many studies approach the problem as an anomaly detection task [57], [100], [101], [102], [103].

a) *Autoencoders*: They learn a compressed (low-dimensional) feature representation, which is called latent vector, to reconstruct the input. Autoencoders are trained by minimizing the pixelwise error between input and reconstructed images to obtain a better reconstruction. Mujeeb et al. [101] employed linear autoencoders to learn from normal solder joints. The degree of anomaly in the test images is evaluated by calculating the Euclidean distance between normal and test images' feature vectors. The test images close to normal are evaluated as normal, and far ones are defective.

b) *Generative models*: They learn the data distribution to generate new samples unlike autoencoders, which are deterministic. Besides, autoencoders' latent space is discontinuous. This limits reconstruction. Most frequently used generative models are latent variable models, as given in Fig. 10. These models learn to generate samples from latent variables. Goto et al. [103] modeled normal solder joints on X-ray

images by adversarial autoencoders (AAEs) that have a more filled latent space than autoencoders. AAE turns the autoencoder into a generative model by converting data distribution to prior distribution over the latent variables and mapping imposed prior to the data distribution [104]. It is a combination of variational autoencoders (VAEs) and generative adversarial networks (GANs). The degree of anomaly is calculated by the Mahalanobis distance. However, adversarial training is not stable and it can cause mode collapse. Hence, Li et al. [86] employed a least-squares GAN, which provides a more stable loss function, to model normal IC solder joints. The pixel difference between normal and defective solder joint is used as an anomaly score.

On the other hand, resolving underlying factor of variations in the data is important to learn a better representation. Ulger et al. [57] modeled both IC and passive SMD solder joints with a beta-variational autoencoder (beta-VAE) architecture to learn a disentangled representation. VAE imposes a prior distribution over the latent vector by minimizing KL divergence. Both gradient [57] and activation-based methods [57], [100] are used to assign anomaly score to each of the test samples.

To sum up, unsupervised generative models do not require defective solder joints, which are already scarce in this task, during training at the expense of accuracy compared to the supervised methods.

3) *Others*: Solorzano and Tsai [105] proposed using reinforcement learning (RL) to register reference and test images, which is helpful in the inspection task. In RL, there is a learner that takes actions in an environment by trial-and-error and feedback received from the environment to maximize

rewards [106]. In this context, the RL model is trained with a reference and a misaligned test image to learn the required geometric transformations to register these two images. Registration with RL is shown to be more effective and robust against environmental conditions than template, feature matching, and pure CNN.

Zheng et al. [102] experienced that solder paste printing can be prone to errors if one treats it as a single task since there are many different electronic components and solder pads. Therefore, they used multitask learning to conform the difference in solder paste applications. Multitask learning enables gathering knowledge from the other related tasks that can be useful in the current task. Initially, tasks are generated from a set of geometric features of solder paste along with labels based on a tree structure. Then, similar tasks are grouped to transfer knowledge by a clustering algorithm. A model is optimized over a loss function to approximate correct labels. Thereby, solder joint inspection can be carried out in a more general context.

Remarks: Unsupervised models aim to learn a representation without labeled data that is hard to obtain in solder joint classification. Moreover, unsupervised models can learn to model normal solder joints without the need for defective samples in the training data. Their classification accuracy is comparatively lower than supervised learning due to a lack of label supervision. To bridge this gap, multitask learning methods can transfer knowledge from other domains where data are abundant. However, data distributions of different datasets are generally different. Therefore, how unsupervised learning can improve model generalization on the target task is an open question [107]. If limited labels are available, semisupervised learning utilizes them as prior knowledge that provides better generalization for the target task.

D. 3-D Reconstruction

Solder joint surfaces (solder paste) have 3-D shapes that are regulated by IPC. Hence, assessing solder joint quality by their original shapes is important. Besides, 3-D images give the information that is not available in 2-D, which can be insufficient to distinguish normal and defective samples. In earlier methods, Kim et al. [61] used specular characteristics of solder surface. Tilt angles over solder joints are used for each illumination angle to obtain 3-D shape information from 2-D images; however, this approach is not precise enough since there is a loss of information with the transition between dimensions.

A more prevalent approach is 3-D reconstruction/modeling of solder paste. There are mainly two methods that are used for 3-D reconstruction of solder paste/joints: laser triangulation [109], [110], [111], [112], [113] and phase shift profilometry [114], [115], [116], [117], [118]. Lathrop [109] measured the height, width, and area of a solder paste to assess its printability with laser triangulation, where a point laser source is projected to the solder paste, as shown in Fig. 11(a). The light reflected from the specular solder surface is captured by an optic sensor (camera). By moving the source to the top and bottom of solder paste, we obtain the

height of the solder paste, which appears on the sensor. The downside is the measured time, which takes several hours for each solder paste. In order to speed up the process, Yang et al. [110] proposed reconstructing and measuring the 3-D profile of a solder paste by training an NN with solder paste height measurements taken by a laser system. Similarly, Kuo et al. [111] proposed the 3-D reconstruction of a solder paste with laser scanning and multiple light projections to be able to inspect different solder pad types. An NN is trained with images captured under different light projections. Then, the trained model is used in the test to obtain the height of the solder paste. This modeling approach does not necessitate laser scanning for each solder paste in the test and therefore speeds up the reconstructions tremendously. However, inconsistent reflectivity of PCB background causes inaccurate measurements. Hence, Xiaohui et al. [112] showed that images with low gray values exhibit much noise, which results in inaccurate reconstructions. They propose an image fusion method to remedy this issue and fuse two images captured by a high dynamic range camera with different gray values. This prevents amplification of noise and hence can yield better reconstructions.

Instead of directly using laser scanning for 3-D reconstruction, Li et al. [113] employed relation-shape CNN (RS-CNN) for 3-D reconstruction of laser-scanned solder joints. RS-CNN produces a set of 3-D discrete points that represent the shape of the solder joints. The point cloud representation is more robust against rigid transformation [119], which can be helpful in solder joint inspection. Furthermore, a soft attention-based NN is used to focus on discriminative regions on the point cloud. A weighted cross-entropy loss function, which guides the distribution of the defective samples to a smaller range, is proposed to improve the classification accuracy. Alternatively, Hu et al. [120] used a binocular light detection and ranging (LiDAR) system, which is illumination invariant, to capture 3-D point clouds of solder joints more accurately. This enables more robust inspection for solder joints and does not require the widely used tiered RGB illumination.

Phase shift profilometry is faster and has a lower cost compared to laser triangulation. Phase shift profilometry is based on creating a structured pattern also known as a fringing pattern over the object and phase measurements [114], [121], [122]. A structured light is projected onto the solder paste to create a phase-modulated sinusoidal pattern and viewed by a camera, as shown in Fig. 11(b). When there is a difference between surface heights, shifting occurs in the patterns. Thereby, the phase information allows the 3-D reconstruction of the object by extracting phase by the Fourier transform. In order to recover the sinusoidal pattern closer to its ideal form, Hui and Pang [115] proposed a two-step phase shift profilometry-based 3-D modeling to model solder paste. Two sinusoidal out-of-phase patterns are created over a solder paste instead of one to estimate the background. Yen et al. [116] used an LCD grating to create phase-modulated patterns. Thereby, the reconstruction area can be scaled up. This is especially useful to measure multiple solder paste heights at once. Reconstructed 3-D solder paste's length,

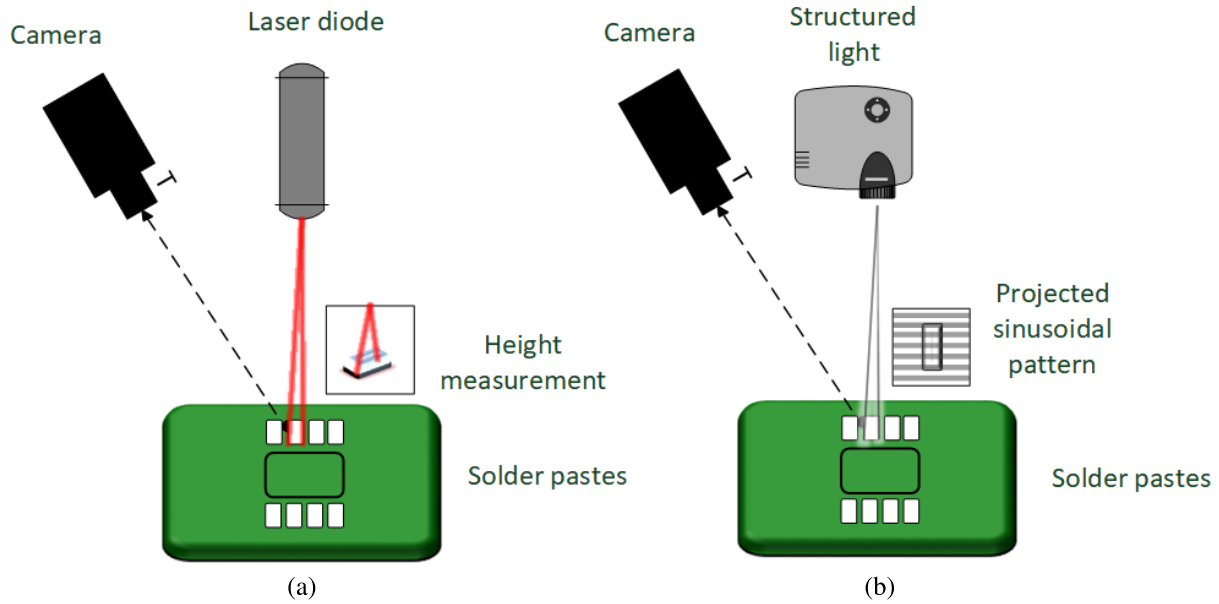


Fig. 11. (a) Laser triangulation and (b) phase shift profilometry for 3-D reconstruction of solder paste (inspired by [118], [121], and [122]).

width, area, volume, aspect ratio, and standard deviation of heights are used as features in inspection [117]. Kuo and Wu [118] showed that reflection coefficients are different for flip-chip board and solder balls, and this results in phase errors and inaccurate measurements as for the laser triangulation. They propose generating color composite fringe by incorporating fringe information with different modulation intensities in each channel. The proposed solution is shown to be more accurate in the 3-D reconstruction of solder balls than standard color fringe.

Remarks: The 3-D reconstruction of solder joints allows taking precise measurements that are compared with IPC standards for the final assessment. This method is reliable and is a prevalent approach in the industry. It is mainly achieved by time-consuming sensor measurements based on laser triangulation and phase shift profilometry. However, high speed is also demanded in the optical inspection of solder joints in a PCB assembly line. This can be alleviated with deep models that can do promising 3-D reconstructions from 2-D images since they do not necessitate time-consuming sensor measurements. Although the presented methods seem to do satisfactory 3-D reconstruction, there is no benchmark study that presents a qualitative comparison. Besides, most of the 3-D representations of solder paste with deep models are constructed with point cloud representation that lacks connectivity information. Since solder errors are rather small and inconspicuous, 3-D representations, such as volumetric and multiview representations, which can preserve fine details of the object [123], may lead to better reconstructions.

VI. DISCUSSION

Many different approaches are proposed over the years, which aim to carry out optical inspection of PCBs. All of these methods have advantages and disadvantages that should be considered when applying them in a real production line.

Reference-based methods are easy to implement, and however, they require a reference (defect-free) version of a PCB under test for comparison. Image subtraction-based comparison suffers from environmental changes such as light and translation. It requires robust registration to compare the corresponding pixels of the reference and defective PCB images. On the other hand, template matching provides a more robust comparison than image subtraction. However, template matching requires a lot of templates to generalize well under test and is computationally expensive for large images.

Traditional ML/CV yields very high accuracy and is shown to detect different types of errors. In the literature, most of the approaches are based on extracting features from solder surfaces by using RGB illumination. Yet, these methods require determining distinctive features, defective samples to extract features, and mostly a specialized RGB illumination.

Deep learning-based methods are also powerful when applied to different PCB inspection tasks. Supervised models can leverage labeled data, which can guide the model to an optimal solution to achieve the highest accuracy. These models require a lot of labeled normal and defective samples. Unsupervised models can inspect by learning only from normal solder joints. They do not require defective solder joints in training, which are hard to obtain, yet they cannot achieve as high accuracy as the supervised models. Furthermore, object detectors enable both localization and classification of the errors from a PCB with a single model. Although these models do not require handcrafted features, they require a lot of data to learn from.

The 3-D reconstruction of solder joints and SMDs yields highly accurate measurements, such as height, for inspection purposes without any loss of information as in 2-D imaging. Consequently, 3-D inspection offers superior accuracy compared to its 2-D counterpart. The measurements of solder pastes are evaluated against IPC-A-610 standards to validate

TABLE VII
PRESENTED METHODS FOR THE SOLDER JOINT INSPECTION, MODEL FAMILIES, AND BASIS BY THEIR USE CASES

Case	Model Basis	Model Family	Methods
Reference and defective PCBs available	Referential approach	Pixelwise comparison Template matching	Image subtraction [27], [30], [33] Normalized cross-correlation [41]–[44]
Hand-crafted features	ML/CV	Color features Geometric features Others	Gray values, percentage of highlight [52], [53], [55], [61], color changes [62] Perimeter, length, height, width [59], [66], Solder area [62], [64], [65], curvature, inertia [67] Wavelet [43], [45], [68], Gabor [49]
Special hardware (Sensors, structured light etc.)	External information	3D reconstruction	Laser triangulation [109]–[112], Point cloud representation [113], [120] Phase shift profilometry [114]–[118], LIDAR [120]
Learning from only normal samples	ML/CV	Others	color space distribution [34], [82], [127], manifold learning [81] PCA [80], FISTA [33]
	Unsupervised learning	Autoencoders Generative models	Linear autoencoders [101] VAE [100], β -VAE [57], LS-GAN [86], AAE [103]
Labelled normal and defective samples	Supervised learning	Pure CNN Sequence modeling FGIC	Dual stream CNN [84], [87], ResNet-50 [88], weighted cross-entropy [89] LSTM [19], [90]–[92] Entropy regularization [98], triplet loss [16], focal loss [96]
Localization and classification with a single step	Object detectors	Region proposal Classification/regression	Faster R-CNN [94], Mask R-CNN [85], FPN [95], [96] YOLO [94], [97], YOLACT [98]

ML/CV machine learning/computer vision

the quality of the solder joint. As a result, 3-D AOI has become the industry norm. Nonetheless, 3-D reconstruction necessitates specialized hardware and requires more time. To mitigate these challenges, deep learning models are integrated to enhance the 3-D reconstruction and reduce reconstruction time.

Table VII provides an overview of the methods proposed in the literature for solder joint inspection, categorized by their use cases, model basis, families, and methodology. Although no study has conducted experiments to compare all the presented approaches for the same task, it can be inferred that the appropriate approach should be selected based on the use case. In terms of accuracy, supervised deep learning and 3-D reconstruction-based methods have the potential to outperform others, provided that sufficient data or time is available, considering the aforementioned factors.

VII. FUTURE DIRECTIONS

In recent years, deep learning has emerged as the primary method for PCB inspection due to its high accuracy. Some approaches even eliminate the need for widely used hemispherical/circular RGB illumination, which can be space-consuming. However, training or deploying these models can be computationally intensive due to the large number of parameters they require. If the computation can be handled efficiently, for example, by using lightweight models, PCB inspection could be performed on mobile devices such as smartphones and augmented reality glasses in the near future. This would enable inspections to be conducted outside laboratory environments and would be beneficial for electronics hobbyists and professionals alike.

Solder joint errors are rather small and inconspicuous, so it is hard to recognize them at a first glance. However, attention-based models such as Vision Transformers (VTs) can help by focusing on these subtle details, making it easier to make

accurate predictions. VT is a recent and powerful type of attention-based model that consists of multiple self-attention layers and an MLP. First, the input image is divided into a sequence of patches with positional information to keep spatial information of the data. The resultant vectors are then fed into the self-attention module, which integrates the information from different layers to produce an output. The output is then passed through an MLP with weights that are optimized to minimize cross entropy during training. Finally, class probabilities are assigned to make the final prediction. VTs have been shown to achieve higher accuracy in image recognition compared to CNN- and RNN-based models, making them a promising option for solder joint inspection.

In the absence of defective solder joints, unsupervised generative models hold promise as an alternative solution. One such successful recent generative model is the diffusion model, which creates a Markov chain using both forward and reverse processes. During the forward pass, the model gradually adds noise to the input image, which is then used to generate new data during the reverse process. Unlike VAE and GAN, diffusion models are stable during training and can produce high-quality and diverse samples. These models can be used to model defect-free solder joints and subsequently detect anomalies in the data.

An alternative research direction for PCB inspection is few-shot learning, which is particularly useful when obtaining labeled data is difficult. This is often the case in PCB inspection, where collecting enough defective PCB errors can be a challenge. Few-shot learning approaches aim to learn from only a few samples, compensating for the lack of supervised information by utilizing prior knowledge. One popular method for acquiring this prior knowledge is meta-learning, which involves learning from other similar tasks to inform the target task. Prior knowledge can be applied to the data, model, or algorithm. For example, one can transfer data from similar

datasets, such as SMD images or PCB patterns to increase the number of samples in training. Models that incorporate prior knowledge, such as generative models or multitask learning, can be selected to constrain the hypothesis space and approximate the ground truth. The algorithm refers to the strategy for finding model parameters that best approximate the model distribution to the ground-truth distribution. By leveraging prior knowledge, the search space for the model parameters that generalize well can be constrained. In addition, meta-learned parameters from related tasks can be useful in the target task [128].

VIII. CONCLUSION

In this survey, we discuss and compare various approaches that have been proposed for PCB inspection, considering their respective advantages and disadvantages. Specifically, we concentrate on the more recent deep learning and 3-D reconstruction-based methods. In addition, we have released a public dataset that includes both normal and defective solder joints, which can serve as a benchmark for comparing different approaches and addressing any gaps in the existing literature.

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Furkan Ulger received the B.S. and M.S. degrees in electrical and electronics engineering from Hacettepe University, Ankara, Turkey, in 2019 and 2022, respectively.

He is currently working at Aselsan Inc. His research interests include computer vision and machine learning.



Seniha Esen Yuksel (Senior Member, IEEE) received the B.S. degree in electrical and electronics engineering from Middle East Technical University, Ankara, Turkey, in 2003, the M.S. degree in electrical and computer engineering from the University of Louisville, Louisville, KY, USA in 2005, and the Ph.D. degree in computer engineering from the University of Florida, Gainesville, FL, USA, in 2011.

She is currently an Associate Professor with the Department of Electrical and Electronics Engineering, Hacettepe University, Ankara. She is also the Director of the Pattern Recognition and Remote Sensing Laboratory (PARRSLAB), Hacettepe University, where she is doing research on machine learning and computer vision with applications in the defense industry.



Dincer Gokcen (Member, IEEE) received the B.S. degree in electrical engineering from Yildiz Technical University, Istanbul, Turkey, in 2005, and the Ph.D. degree in electrical engineering from the University of Houston, Houston, TX, USA, in 2010.

He has been with the National Institute of Standards and Technology, Gaithersburg, MD, USA, GlobalFoundries, Malta, NY, USA, and Aselsan Inc., before joining Hacettepe University, Ankara, Turkey, as a Faculty Member in 2016. His research interests include nanofabrication, microsystems, and sensors.



Atila Yilmaz received the M.Sc. degree in control and information technology from the Control Systems Centre, University of Manchester Institute of Science and Technology, Manchester, U.K., in 1992, and the Ph.D. degree from the Biomedical Engineering Division, University of Sussex, Brighton, U.K., in 1996.

He is currently a Professor with the Department of Electrical and Electronics Engineering, Hacettepe University, Ankara, Turkey. His research interests include artificial intelligence and neural networks in control and biomedical applications, especially for image and biomedical signal processing.